Musicians, Memory, and General Cognitive Ability: A Commentary on Grassi et al. (2025)

Enrico Toffalini and Tommaso Feraco

Department of General Psychology, University of Padua

Author Note

Enrico Toffalini https://orcid.org/0009-0001-4661-8063

Tommaso Feraco https://orcid.org/0000-0002-8920-5330

Correspondence concerning this article should be addressed to Enrico Toffalini, Department of General Psychology, University of Padua, 8 Via Venezia, Padua, PD, Italy, Email: enrico.toffalini@unipd.it

Musicians, Memory, and General Cognitive Ability: A Commentary on Grassi et al. (2025)

Grassi et al. (2025) provide the strongest dataset to date on cognitive differences between musicians and nonmusicians. Their contribution is exemplary in scale, openness, and collaborative rigor, and shows how a literature long constrained by underpowered studies and heterogeneous protocols can be meaningfully advanced. Their core empirical contribution is not in question. Here, we focus on a narrower but theoretically important issue, the level at which the cognitive findings can be most meaningfully interpreted.

The article tests whether musicians outperform nonmusicians not only on music memory, where a large effect is unsurprising, but also on verbal and visuospatial short-term memory, where any differences were less obvious a priori. The discussion appropriately considers several hypotheses, including music-specific transfer, broader cognitive differences, and selection effects. However, its interpretation remains closer to a task-by-task reading of the cognitive results than is warranted by the broader theory of cognitive individual differences.

In intelligence research, the observation of a positive manifold across cognitive-task performances is a pivotal starting point, and mainstream psychometric accounts continue to treat the variance common to different cognitive tests as theoretically fundamental, whether labeled general intelligence or "g" (Coyle, 2021; Coyle & Greiff, 2023; Spearman, 1904). Although the nature of "g" may remain somewhat elusive, its neurobiological and genetic roots are increasingly being recognized (Deary et al., 2022). Latent variables remains central to mainstream psychometric theories of cognitive abilities and, more broadly, to major theories of individual differences in diverse areas like personality and clinical psychology (Wright, 2020). In all these cases, the implication is the same: observed tasks or items are not psychologically independent just because they are superficially different. When several indicators covary, the first question is whether group differences are expressed at the level of shared or nonshared variance, rather than whether each observed score should be treated separately as evidence for a distinct, specific difference.

This matters directly here. Grassi et al. measured not only memory tasks but also Raven,

Wechsler Vocabulary, and n-back performance. These scores were then entered into mixed-effects models as control variables, implicitly treating them as direct causes of memory performance or, more pragmatically, as variables to be conditioned on when asking whether musicians still outperform nonmusicians at the same level of other cognitive abilities (yet without a measurement model, such strategy carries a substantial risk of spurious conclusions, Westfall & Yarkoni, 2016). In their discussion, the authors themselves noted that the broadly positive pattern across cognitive variables may reflect a more general cognitive advantage, with the clearest and largest difference emerging when the task is closest to musical expertise. We believe this should not remain a side note, but should instead guide a more parsimonious and theoretically meaningful interpretation of the findings.

We reanalyzed the data following the mainstream view that common variance among cognitive measures (interpreted as g) should be modeled first. The cognitive tasks, however, do not appear to have been selected with an explicit measurement model in mind. We therefore fit a one-factor model to the six cognitive measures, Vocabulary, Raven, n-back, digit span, spatial span, and melody span, while allowing for a small number of residual correlations. The positive manifold emerged, but was not very strong: after controlling for group, the average intercorrelation among observed scores was about .20, lower than is often observed in standard intelligence batteries. This may reflect the limited reliability of some indicators, the relatively restricted and homogeneous sample, or the fact that the battery was not originally designed as a psychometric measure of general cognitive ability. Nonetheless, it requires being modelled.

Musicians and nonmusicians were perfectly or nearly perfectly matched on age, sex/gender, and years of education, whereas they differed in socioeconomic status (SES), with Cohen's d = 0.34 in favor of musicians. We therefore adjusted only for SES when testing group effects, to avoid unnecessary model complexity. It should be recognized, however, that the causal status of SES here remains unclear. Group effects were examined in two ways: first, through multigroup confirmatory factor analyses (MG-CFA) within a standard measurement-invariance framework, leading to a test of equality of the specific observed intercepts and of the latent mean of g. Second,

via multilevel structural equation models (SEM), in which group predicted common versus task-specific variance, while country-level clustering was accounted for through random intercepts on the observed variables. Because `lavaan` did not allow simultaneous multigroup and multilevel estimation in this case, these approaches had to be implemented separately. SES was included as a covariate when testing group effects. Model fit was evaluated using RMSEA, CFI, and BIC. Final interpretation privileged models that combined parsimony (lowest BIC) with acceptable fit (RMSEA < 0.05, CFI > 0.95). For model coefficients, we considered statistical significance at $p < .05$. All latent-variable models were fitted in R using the `lavaan` package (Rosseel, 2012).

The multigroup invariance analyses supported configural invariance, RMSEA = .027 and CFI = .986, and metric invariance, RMSEA = .029 and CFI = .979, with BIC improving from 19525.72 to 19493.60. Full scalar invariance fit poorly, but partial scalar invariance was supported once the intercept of Melody span was freed, RMSEA = .035, CFI = .966, BIC = 19478.61. Adding the equality constraint on the SES -> g regression did not materially worsen fit, RMSEA = .035, CFI = .966, BIC = 19472.67, and this was retained as the final preferred multigroup representation Figure 1. A still more constrained model with equal residual covariances yielded a trivially smaller BIC (19472.34) but worse approximate fit, RMSEA = .044 and CFI = .940, so it was not preferred for final interpretation. By contrast, strict invariance was not supported, RMSEA = .062, CFI = .871, BIC = 19501.56. Constraining the latent mean of g to equality also worsened fit, RMSEA = .057, CFI = .904, BIC = 19509.60, indicating that the groups differed at the latent level. In the preferred partial-scalar model, the estimated latent mean difference was 0.51 latent standard deviations in favor of musicians. Thus, the multigroup results support a parsimonious account in which musician status is associated primarily with higher general cognitive variance, while Melody span also shows clear intercept noninvariance, consistent with an additional task-specific advantage. SES positively predicted g in both groups, with very similar standardized coefficients, about .22 in nonmusicians and .20 in musicians.

The multilevel SEM reanalysis pointed in the same general direction, while also refining the decomposition of group differences. Among the multilevel models considered, the preferred

solution was the targeted hybrid model in which group predicted the latent factor g and, in addition, retained direct residual paths only to Wechsler Vocabulary and Melody span, RMSEA = .027, CFI = .979, BIC = 19133.55. At the within-country level, which is the only level we interpreted substantively, musician status predicted higher latent g after adjustment for SES, with a standardized effect of about .42. SES also positively predicted latent g, standardized coefficient about .23. Beyond this general effect, the model suggested a small residual advantage for Wechsler Vocabulary, standardized effect about .08, and a clearly larger residual advantage for Melody span, standardized effect about .43. This complements the multigroup results by again suggesting that most of the musician advantage is captured at the latent level, while leaving a more specific residual difference where theory would most expect it, namely in the music-relevant indicator, plus a smaller vocabulary-specific deviation.

Both our reanalyses suggest that the musician advantage is expressed primarily at the latent level, with a clear additional specificity for Melody span and, possibly, a smaller residual effect for Vocabulary. This is not a denial of the original empirical findings, but a more parsimonious account within a methodological framework that is better aligned with the still-dominant mainstream on individual differences in cognitive abilities. The point is not just to adjudicate between “significant” and “nonsignificant” group differences, but to clarify the most appropriate explanatory framework: whether a set of observed scores should be interpreted as a collection of separate outcomes, with their covariance treated as a nuisance to be removed through pragmatic analytic choices, or whether the overall pattern is best understood as the surface expression of true common factor(s). This matters beyond the present article. Large collaborative datasets are a major advance for the field, and their value can be strengthened even further when they are paired with designs that more fully reflect the hierarchical structure of cognitive abilities (McGrew, 2009). When indicators are positively manifolded, modeling them jointly can help reveal whether apparently distinct group differences are better interpreted as manifestations of a broader common factor. Therefore, advances in data collection and research practices can and should be matched by equally strong advances in measurement modeling, yielding interpretations

that are both more integrated and more theoretically meaningful.

References

Coyle, T. R. (2021). *Defining and measuring intelligence: The psychometrics and neuroscience of g* (A. K. Barbey, R. J. Haier, & S. Karama, Eds.; pp. 3–25). Cambridge University Press. https://doi.org/10.1017/9781108635462.003

Coyle, T. R., & Greiff, S. (2023). Carbon is to life as *g* is to _____: A review of the contributions to the special issue on specific abilities in intelligence. *Intelligence*, *101*, 101786. https://doi.org/10.1016/j.intell.2023.101786

Deary, I. J., Cox, S. R., & Hill, W. D. (2022). Genetic variation, brain, and intelligence differences. *Molecular Psychiatry*, *27*(1), 335–353. https://doi.org/10.1038/s41380-021-01027-y

Grassi, M., Talamini, F., Altoè, G., Brattico, E., Caclin, A., Carretti, B., Drai-Zerbib, V., Ferreri, L., Gambarota, F., Grahn, J., Guiotto Nai Fovino, L., Roccato, M., Rodriguez-Fornells, A., Swaminathan, S., Tillmann, B., Vuust, P., Wilbiks, J., Zentner, M., Aguilar, K., … Zappa, A. (2025). Do Musicians Have Better Short-Term Memory Than Nonmusicians? A Multilab Study. *Advances in Methods and Practices in Psychological Science*, *8*(4), 25152459251379432. https://doi.org/10.1177/25152459251379432

McGrew, K. S. (2009). CHC theory and the human cognitive abilities project: Standing on the shoulders of the giants of psychometric intelligence research. *Intelligence*, *37*(1), 1–10. https://doi.org/10.1016/j.intell.2008.08.004

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, *48*(2), 1–36. https://doi.org/10.18637/jss.v048.i02

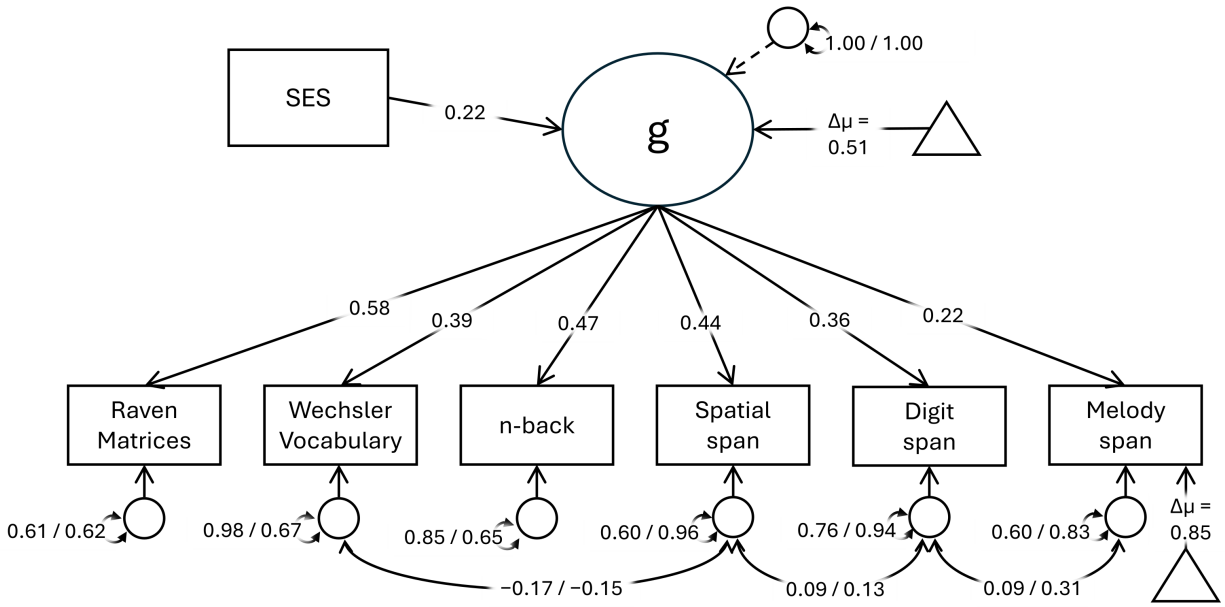
Spearman, C. (1904). General intelligence, objectively determined and measured. *The American Journal of Psychology*, *15*(2), 201–292. https://doi.org/10.2307/1412107

Westfall, J., & Yarkoni, T. (2016). Statistically Controlling for Confounding Constructs Is Harder than You Think. *PLOS ONE*, *11*(3), e0152719. https://doi.org/10.1371/journal.pone.0152719

Wright, A. G. C. (2020). *Latent variable models in clinical psychology* (A. G. C. Wright & M. N.

Hallquist, Eds.; pp. 66–79). Cambridge University Press.

https://doi.org/10.1017/9781316995808.009

Figure 1

Final multigroup model representation. Dual values separated by a slash denote the non-musician (left) vs musician (right) coefficients. Triangles (for intercepts) indicate where group differences in mean values emerged. Unstandardized coefficients were reported because standardized estimates are variance-dependent and may differ across groups due to residual variance differences rather than true parameter differences. All reported cofficients are statistically significant with $p < 0.05$. SES = Socio-Economic Status.